

U.S. Billion-Dollar Disasters — Market Cross-Reference

Generated: 2026-06-18 20:21 | Source: NOAA NCEI (1980–2024) · Thor Storehouse | Cross-referenced: WTI · GASREGCOVW · INDPRO
| 400 events · 377 with WTI data

Each disaster event is cross-referenced against WTI crude oil price (± 5 days), retail gasoline price (± 10 days), and the Federal Reserve Industrial Production Index (± 30 days) using data archived in Thor Storehouse from FRED. All cost figures CPI-adjusted to 2024 dollars. No opinion. No forecasting. #FollowTheData

Executive Summary

Across 377 eligible disaster events since 1986, average WTI crude oil movement was only **+0.06%** — suggesting weather disasters alone have not historically produced a reliable directional signal for oil prices.

\$2,912B

CPI-ADJUSTED LOSSES

16,941

DEATHS

+0.06%

AVG WTI REACTION

Despite enormous human and economic costs — markets showed almost no consistent reaction.

Key Finding: Major U.S. weather disasters historically show little consistent relationship with WTI crude oil prices, retail gasoline prices, or industrial production. WTI averaged **+0.06%** — effectively flat — across 377 qualifying events. Oil rose after 56 events and fell after 49, with no directional pattern by category, cost, or region. Retail gasoline averaged +0.39% and industrial production averaged +0.06% — both effectively flat. Market movements reflect many simultaneous macroeconomic factors. Disaster causation cannot be inferred from short-window correlation.

TOTAL EVENTS

400

NOAA NCEI · 1980–2024 ·
CPI-adjusted

TOTAL CPI-ADJ COST

\$2,912.1B

2024 dollars · direct losses

TOTAL DEATHS

16,941

Direct deaths attributable to
event

WTI AVG REACTION

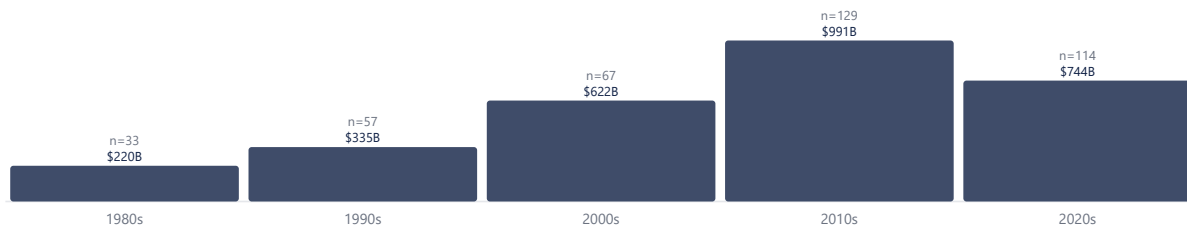
+0.06%

≈ zero · 377 qualifying
events

Rose: 56 · Fell: 49 · Flat: 272

COST AND FREQUENCY BY DECADE

CPI-adjusted · NOAA NCEI · Storehouse



BY DISASTER CATEGORY

Category	Events	Total Cost	Deaths
Tropical Cyclone	67	\$1543.0B	7,211
Severe Storm	200	\$508.5B	2,145
Drought	32	\$367.8B	4,658
Flooding	45	\$203.0B	765
Wildfire	23	\$148.0B	537
Winter Storm	24	\$104.4B	1,463
Freeze	9	\$37.3B	162

MARKET REACTION SUMMARY

WTI — events before 1986 (no data)	23
WTI — events after 1986 (eligible)	377
WTI — matched within ±5 days	377
WTI — no data in window	0
WTI — avg % change	+0.06%
WTI — rose / fell / flat	56 / 49 / 272
Gas — events matched	363
Gas — avg % change	+0.39%
INDPRO — events matched	400
INDPRO — avg % change	+0.06%

Why only 377 of 400 events have WTI data:
FRED's daily WTI crude oil series (DCOILWTICO) begins January 2, 1986. 23 events occurred before that date and cannot be matched — this is a source data limitation, not a reporting gap. Of the 377 events after 1986, 0 had no WTI observation within ±5 trading days (typically holiday periods or data gaps). The 377 matched events are the complete eligible universe.

TOP 25 COSTLIEST EVENTS CPI-adjusted 2024 dollars

Year	Event	Category	Cost (adj.)
2005	Hurricane Katrina (August 2005)	Tropical Cyclone	\$201.3B

Year	Event	Category	Cost (adj.)	Deaths
2017	Hurricane Harvey (August 2017)	Tropical Cyclone	\$160.0B	119
2022	Hurricane Ian (September 2022)	Tropical Cyclone	\$119.6B	154
2017	Hurricane Maria (September 2017)	Tropical Cyclone	\$115.2B	2991
2012	Hurricane Sandy (October 2012)	Tropical Cyclone	\$88.5B	233
2021	Hurricane Ida (August 2021)	Tropical Cyclone	\$84.6B	103
2024	Hurricane Helene (September 2024)	Tropical Cyclone	\$78.7B	233
2017	Hurricane Irma (September 2017)	Tropical Cyclone	\$64.0B	11
1992	Hurricane Andrew (August 1992)	Tropical Cyclone	\$60.5B	65
1988	U.S. Drought/Heat Wave (Summer 1988)	Drought	\$54.6B	1700
1993	Midwest Flooding (Summer 1993)	Flooding	\$46.3B	50
2008	Hurricane Ike (September 2008)	Tropical Cyclone	\$43.2B	12
2012	U.S. Drought/Heat Wave (2012)	Drought	\$41.7B	1200
1980	Central/Eastern Drought/Heat Wave (Summer-Fall 1980)	Drought	\$40.7B	1700
2024	Hurricane Milton (October 2024)	Tropical Cyclone	\$34.2B	29
2004	Hurricane Ivan (September 2004)	Tropical Cyclone	\$34.0B	11
2018	Hurricane Michael (October 2018)	Tropical Cyclone	\$31.2B	22
2005	Hurricane Wilma (October 2005)	Tropical Cyclone	\$30.0B	11
2018	Hurricane Florence (September 2018)	Tropical Cyclone	\$30.0B	14
2018	Western Wildfires, California Firestorm (Summer-Fall 2018)	Wildfire	\$30.0B	33
2005	Hurricane Rita (September 2005)	Tropical Cyclone	\$29.4B	26
2020	Hurricane Laura (August 2020)	Tropical Cyclone	\$28.1B	12
2021	Northwest, Central, Eastern Winter Storm and Cold Wave (February 2021)	Winter Storm	\$27.2B	12
2004	Hurricane Charley (August 2004)	Tropical Cyclone	\$26.7B	48
2022	Western/Central Drought and Heat Wave (2022)	Drought	\$23.5B	1200

TOP 25 DEADLIEST EVENTS Direct deaths

Year	Event	Category	Cost (adj.)	Deaths
2017	Hurricane Maria (September 2017)	Tropical Cyclone	\$115.2B	2991
2005	Hurricane Katrina (August 2005)	Tropical Cyclone	\$201.3B	1833
1980	Central/Eastern Drought/Heat Wave (Summer-Fall 1980)	Drought	\$40.7B	1700
1995	Central, Southern and Northeast Drought/Heat Wave (September 1995)	Drought	\$2.0B	1600

Year	Event	Category	Cost (adj.)	Impact
1999	Eastern Drought/Heat Wave (Summer 1999)	Drought	\$4.8B	1
1988	U.S. Drought/Heat Wave (Summer 1988)	Drought	\$54.6B	1
2011	Southeast/Ohio Valley/Midwest Tornadoes (April 2011)	Severe Storm	\$14.3B	1
1993	East Coast Blizzard and Severe Weather (March 1993)	Winter Storm	\$12.2B	1
2021	Northwest, Central, Eastern Winter Storm and Cold Wave (February 2021)	Winter Storm	\$27.2B	1
2023	Southern/Midwestern Drought and Heatwave (Spring-Fall 2023)	Drought	\$14.8B	1
2021	Western Drought and Heat Wave (2021)	Drought	\$10.2B	1
2024	Hurricane Helene (September 2024)	Tropical Cyclone	\$78.7B	1
1998	Southern Drought and Heat Wave (Summer 1998)	Drought	\$6.9B	1
1996	Blizzard/Floods (January 1996)	Winter Storm	\$6.1B	1
2011	Midwest/Southeast Tornadoes (May 2011)	Severe Storm	\$12.7B	1
2012	Hurricane Sandy (October 2012)	Tropical Cyclone	\$88.5B	1
2022	Hurricane Ian (September 2022)	Tropical Cyclone	\$119.6B	1
1983	Freeze/Cold Wave (December 1983)	Freeze	\$6.5B	1
1985	Winter Storm, Cold Wave (January 1985)	Winter Storm	\$2.5B	1
2000	Western/Central/Southeast Drought/Heat Wave (Spring-Fall 2000)	Drought	\$9.3B	1
2022	Western/Central Drought and Heat Wave (2022)	Drought	\$23.5B	1
2024	Southern/Eastern/Northwestern Drought and Heat Wave (2024)	Drought	\$5.4B	1
1997	Western/Eastern Severe Weather and Flooding (Winter-Spring 1998)	Severe Storm	\$2.0B	1
2012	U.S. Drought/Heat Wave (2012)	Drought	\$41.7B	1
2005	Hurricane Rita (September 2005)	Tropical Cyclone	\$29.4B	1

LARGEST WTI OIL PRICE INCREASES AFTER EVENT ±5 trading days · FRED DCOILWTICO

Year	Event	Category	Cost (adj.)	Impact
1998	Central and Eastern Severe Storms and Flooding (June 1998)	Severe Storm	\$1.7B	1
2020	Tropical Storm Eta (November 2020)	Tropical Cyclone	\$1.8B	1
2023	Southern/Midwestern Drought and Heatwave (Spring-Fall 2023)	Drought	\$14.8B	1
1991	Hurricane Bob (August 1991)	Tropical Cyclone	\$3.5B	1
2012	U.S. Drought/Heat Wave (2012)	Drought	\$41.7B	1
2008	U.S. Drought (2008)	Drought	\$10.5B	1
2020	Central and Eastern Severe Weather (May 2020)	Severe Storm	\$2.6B	1

Year	Event	Category	Cost (adj.)	Deaths
2009	Southwest/Great Plains Drought (2009)	Drought	\$5.2B	0
2006	Numerous Wildfires (2006)	Wildfire	\$2.3B	0
1992	Hurricane Andrew (August 1992)	Tropical Cyclone	\$60.5B	0
2024	Central Tornado Outbreak (May 2024)	Severe Storm	\$3.5B	0
2024	Texas Hail Storms (May 2024)	Severe Storm	\$2.3B	0
1989	Florida Freeze (December 1989)	Freeze	\$5.2B	0
2003	Severe Storms/Tornadoes (May 2003)	Severe Storm	\$7.1B	0
2023	Central Severe Weather (May 2023)	Severe Storm	\$2.2B	0
2024	Upper Midwest Flooding (June 2024)	Flooding	\$1.1B	0
2004	Hurricane Ivan (September 2004)	Tropical Cyclone	\$34.0B	0
1999	Central and Eastern Winter Storm (January 1999)	Winter Storm	\$2.0B	0
2020	Western Wildfires - California, Oregon, Washington Firestorms (Fall 2020)	Wildfire	\$19.9B	0
2016	North/Central Texas Hail Storm (April 2016)	Severe Storm	\$4.6B	0
2015	Central and Eastern Winter storm, Cold Wave (February 2015)	Winter Storm	\$4.0B	0
2006	Northeast Flooding (June 2006)	Flooding	\$2.4B	0
2023	East Coast Storm and Flooding (December 2023)	Flooding	\$1.3B	0
2021	Southern Tornadoes and Southeast Severe Weather (May 2021)	Severe Storm	\$1.5B	0
1996	Blizzard/Floods (January 1996)	Winter Storm	\$6.1B	0

Largest observed WTI increases within 5 days of event start. These events co-occurred with oil price rises — not necessarily caused them.

LARGEST WTI OIL PRICE DECREASES AFTER EVENT

±5 trading days · FRED DCOILWTICO

Year	Event	Category	Cost (adj.)	Deaths
2003	California Wildfires (Fall 2003)	Wildfire	\$6.6B	0
1998	Tropical Storm Frances (September 1998)	Tropical Cyclone	\$1.4B	0
2008	Hurricane Gustav (September 2008)	Tropical Cyclone	\$8.6B	0
2008	U.S. Wildfires (Fall 2008)	Wildfire	\$1.8B	0
2023	Southern Severe Weather (June 2023)	Severe Storm	\$4.2B	0
2002	Western Fire Season (Fall 2002)	Wildfire	\$2.3B	0
1986	Southeast Drought/Heat Wave (Summer 1986)	Drought	\$5.2B	1
2015	Texas Tornadoes and Midwest Flooding (December 2015)	Severe Storm	\$2.6B	0
2014	Western Drought (2014)	Drought	\$5.3B	0

Year	Event	Category	Cost (adj.)	Deaths
2009	Georgia Flooding (September 2009)	Flooding	\$1.3B	
2019	Southeast, Ohio Valley and Northeast Severe Weather (February 2019)	Severe Storm	\$1.5B	
2015	Texas and Oklahoma Flooding and Severe Weather (May 2015)	Flooding	\$3.4B	
2016	Plains Tornadoes and Central Severe Weather (May 2016)	Severe Storm	\$2.3B	
2020	Southeast and Eastern Tornado Outbreak (April 2020)	Severe Storm	\$4.2B	
2011	Midwest/Southeast Tornadoes (May 2011)	Severe Storm	\$12.7B	1
2023	Central and Southern Severe Weather (April 2023)	Severe Storm	\$1.4B	
2021	Western Drought and Heat Wave (2021)	Drought	\$10.2B	2
2003	Western/Central Drought/Heat Wave (Spring-Fall 2003)	Drought	\$8.7B	
1995	California Flooding (January-March 1995)	Flooding	\$5.2B	
2024	Southern/Eastern/Northwestern Drought and Heat Wave (2024)	Drought	\$5.4B	1
1992	Southeast Severe Weather (November 1992)	Severe Storm	\$1.5B	
2016	Houston Flooding (April 2016)	Flooding	\$3.6B	
1998	Northern Plains and Great Lakes Derecho, Tornadoes (June 1998)	Severe Storm	\$2.2B	
2015	Western Drought (2015)	Drought	\$6.0B	
1997	West Coast Flooding (December 1996-January 1997)	Flooding	\$6.0B	

LARGEST RETAIL GAS PRICE INCREASES AFTER EVENT

±10 days · FRED GASREGCOVW

Year	Event	Category	Cost (adj.)
2009	Central Derecho and Tornadoes (May 2009)	Severe Storm	\$1.3B
2008	Hurricane Ike (September 2008)	Tropical Cyclone	\$43.2B
2016	Texas and Louisiana Flooding (March 2016)	Flooding	\$3.1B
2000	Western/Central/Southeast Drought/Heat Wave (Spring-Fall 2000)	Drought	\$9.3B
2000	Western Fire Season (Spring-Summer 2000)	Wildfire	\$2.0B
2020	South, Central and Eastern Severe Weather (May 2020)	Severe Storm	\$1.9B
2009	Southwest/Great Plains Drought (2009)	Drought	\$5.2B
2023	North Central and Eastern Severe Weather (July 2023)	Severe Storm	\$1.5B
2005	Hurricane Dennis (July 2005)	Tropical Cyclone	\$4.0B
2009	Midwest/Southeast Tornadoes (March 2009)	Severe Storm	\$2.4B
2019	Tropical Storm Imelda (September 2019)	Tropical Cyclone	\$6.2B
2016	South/Southeast Tornadoes (April 2016)	Severe Storm	\$3.2B

Year	Event	Category	Cost (adj.)
2007	Spring Freeze (April 2007)	Freeze	\$3.2B
2001	Midwest/Ohio Valley Hail and Tornadoes (April 2001)	Severe Storm	\$5.5B
2008	Midwest Tornadoes and Severe Weather (May 2008)	Severe Storm	\$4.4B
2011	Southern Plains/Southwest Drought and Heat Wave (Spring-Summer 2011)	Drought	\$17.1B
2005	Midwest Drought (Spring-Summer 2005)	Drought	\$2.4B
2022	North Central Severe Weather (May 2022)	Severe Storm	\$2.9B
2006	Midwest/Plains/Southeast Drought (Spring-Summer 2006)	Drought	\$9.5B
2006	Midwest Tornadoes (April 2006)	Severe Storm	\$3.8B
2006	Midwest/Southeast Tornadoes (April 6-8, 2006)	Severe Storm	\$2.5B
2016	Houston Flooding (April 2016)	Flooding	\$3.6B
2019	Southeast, Ohio Valley and Northeast Severe Weather (February 2019)	Severe Storm	\$1.5B
2015	Central and Eastern Winter storm, Cold Wave (February 2015)	Winter Storm	\$4.0B
2015	South/Southeast Severe Weather (April 2015)	Severe Storm	\$1.7B

Data Sources: Disaster events: NOAA NCEI U.S. Billion-Dollar Weather and Climate Disasters (v21.21, final 1980–2024, archived May 2025). WTI crude oil: FRED DCOILWTICO (starts 1986-01-02). Retail gasoline: FRED GASREGCOVW. Industrial production: FRED INDPRO. All time-series data archived in Thor Storehouse.

Methodology: Event begin date used as anchor. WTI: nearest trading day observation before and after within ±5 days. Gas: nearest weekly EIA observation within ±10 days. INDPRO: prior and following monthly readings within ±30 days. Pre-1986 events excluded from WTI analysis — FRED daily series not available. Market movements reflect many simultaneous macroeconomic factors. Disaster causation cannot be inferred from short-window correlation. No opinion. No forecasting. #FollowTheData

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